MIL-P-46133A(MR)
5 August 1974
SUPERSEDING
MIL-P-46133(MR)
30 August 1968

MILITARY SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL, POLY(ARYL SULFONE ETHER) RESIN, THERMOPLASTIC

This specification is approved for use by all departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of thermoplastic resin consisting of phenyl and biphenyl units in a basic polymer structure. The material shall be a general purpose type, opaque beige in color, and in the form of pellets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9330

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

D 149	- Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
D 150	- A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials
D 618	- Conditioning Plastics and Electrical Insulating Materials for Testing
D 638	- Tensile Properties of Plastics
D 648	- Deflection Temperature of Plastics Under Flexural Load
D 792	- Specific Gravity and Density of Plastics by Displacement
D 2240	- Indentation Hardness of Rubber and Plastics by Means of a Durometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order, Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The material shall be a virgin poly(aryl sulfone ether) thermoplastic resin with a basic polymer structure consisting of phenyl and biphenyl units linked by oxygen and sulfone groups. The material shall not contain reinforcing agents such as glass fibers.

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3.2 Property values. The material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.

Table I. Property values

Property	Value
Specific gravity	1.36 ± 0.02
Tensile strength, psi, min.	12,000
Tensile elongation at yield, percent	13 ± 3
Tensile elongation at break, percent	17 ± 6
Heat deflection temperature, min. °C (°F)	271 (520)
Hardness, type D durometer	82 ± 5
Dielectric constant, 60 Hz, max.	4.0
Dissipation factor, 60 Hz, max.	0.005
Dielectric strength, 1/8 inch thick specimen, volts per mil, min.	275

3.3 Color. Unless otherwise specified, the color shall be opaque beige (see 6.2).

3.4 Form. The form shall be pellets. When required by the procuring activity, the form shall be of the size and shape specified (see 6.2).

3.5 Workmanship. The material shall be free from contamination. The color and form of the material shall be uniform within each container and from container to container within each lot (see 4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of this spedification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification will govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table II shall be approximately one pound.

Table II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship	When required, form size not as specified. Not uniform in form. Not uniform in color. Not clean, presence of foreign material.

4.2.2.1.2 Examination of the preparation for delivery. An examination shall be made in accordance with table III to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table III. Examination of preparation for delivery

Examine	Defects
Packing	Not level specified; not in accordance with contract requirements. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted container.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Testing. The molding material shall be tested for the applicable characteristics listed in table I. Testing shall be in accordance with the test methods specified herein for each lot submitted for inspection. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 100 pounds of material. The sample unit of product shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be the averaged results of the specimens, unless only one specimen is required for testing.

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4.2.3.1 Classification of tests. All tests shall be classified, as follows:

a. Lot acceptance tests (see 4.2.3.2).
b. Periodic lot check tests (see 4.2.3.3).

4.2.3.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot check tests are required. Lot acceptance tests shall consist of testing for tensile strength, tensile elongation at yield, tensile elongation at break, and hardness.

4.2.3.3 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring agency (see 6.2). If not specified by the procuring agency, periodic lot check tests shall be repeated at least once every year. Periodic lot check tests shall consist of all tests specified in table I. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot.

4.3 Test methods.

4.3.1 Preparation of specimens. Unless otherwise specified, test specimens shall be prepared by injection molding.

4.3.2 Test conditions. Unless otherwise specified, all tests shall be performed at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity. Specimens shall be conditioned in accordance with procedure A of ASTM D 618.

4.3.3 Specific gravity. One specimen shall be tested in accordance with method A-1 or A-2 of ASTM D 792 using the test specimen size specified in this method.

4.3.4 Tensile strength, elongation at yield, and elongation at break. Five specimens, each 0.125 ± 0.005 inches thick, shall be tested in accordance with ASTM D 638, using Speed B.

4.3.5 Deflection temperature under load. Three specimens shall be tested in accordance with ASTM D 648 using a fiber stress of 264 psi. Length and width of test specimens shall be as specified in D 648 and thickness shall be 0.125 ± 0.015 inch. A circulating air oven should be employed rather than a liquid heat transfer medium because of the high temperatures required.

4.3.6 Hardness. Five measurements shall be made in accordance with ASTM D 2240. The determinations shall be made using the type D durometer specified in ASTM D 2240.

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4.3.7 Dielectric constant and dissipation factor. Three specimens shall be tested in accordance with ASTM D 150, using any electrode system specified in this test method.

4.3.8 Dielectric strength. Three specimens, each 1/8 ± 0.0125 inches in thickness shall be tested in accordance with ASTM D 149, using the short time test.

5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds, maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

In-so-far as practical, containers shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed and strapped in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

In-so-far as practical, containers shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed and strapped in accordance with applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

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5.1.3 Level C. The material shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The thermoplastic material is intended for use in applications requiring good mechanical and electrical properties at 500°F (260°C). Strength is retained at cryogenic temperatures making possible applications requiring a wide temperature range. The material is resistant to acids, bases, hydrocarbon oils and most common solvents. Stability in vacuum is good and resistance to gamma radiation is excellent.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Color, if different (see 3.3).
c. Specific pellet size, if required (see 3.4).
d. Periodic lot check tests, if required (see 4.2.3.3).
e. Level of packing required (see 5.1).

6.3 Cross reference. Type II, form A and color 2 of MIL-P-46133(MR) have been deleted because they are not available and not applicable.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review interest:
Army - EL, ME, AT
DSA - GS

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User interest:
Army - AV

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION	MIL-P-46133A(MR), Plastic Molding and Extrusion Material, Poly (ARYL Sulfone Ether) Resin, Thermoplastic
ORGANIZATION	CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED / DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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